

Harvester Network Best Practice

Replace Ethernet NICs

You may want to replace the Ethernet NICs of a bare-metal node in a Harvester cluster for various reasons, including the following:

- Malfunction or damage
- Insufficient hardware capacity
- Missing features

You can follow the steps below and run them in each node step by step.

Pre-Replacement Checks

1. Verify that the installed Harvester version supports the new NICs.
2. Test the new NICs in non-production environment.
3. On the **Virtual Machines** screen of the **Harvester UI**, verify that the status of all VMs is either **Running** or **Stopped**.
4. On the embedded **Longhorn dashboard**, verify that the status of all Longhorn volumes is **Healthy**.
5. (Optional) On the **Harvester Support** screen, generate a **support bundle** for comparison purposes.

Collect Information

Before any action is taken, it is important to collect the current network information and status.

- Harvester network configuration: By default, Harvester creates a bond interface named `mgmt-bx` for the management network. On top of that is a bridge interface named `mgmt-brx`, which may optionally use a VLAN. Each cluster network also has one new bond interface. You can view the current connection details using the `nmcli` tool.

Example

```
# nmcli
mgmt-br 2021/ connected to vlan-mgmt
"mgmt-br 2021/"
vlan, 8C:89:01:80:C2:F5, sw, mtu 1500
ip4 on vlan1
inet4 10.115.55.50/21
metric 10.115.55.0/21 metric 400
route4 default via 10.115.55.254 metric 400

...

mgmt-bo: connected to bond-mgmt
"mgmt-bo/"
bond, 8C:89:01:80:C2:F5, sw, mtu 1500
master mgmt-br

mgmt-br: connected to bridge-mgmt
"mgmt-br/"
bridge, 8C:89:01:80:C2:F5, sw, mtu 1500

ensd0: connected to bond-slave-ensd0
"ensd0: 8025505 01210101"
ethernet (Lan8), 8C:89:01:80:C2:F5, hw, xriov, mtu 1500
master mgmt-bo

...
```

- Physical NICs: You can use the command `ip link` to retrieve related information, including the state of each NIC and the corresponding master (if applicable).

Example

```
# ip link | grep master -i
2: ensd: BRIDGEPORT_MLTS2021_SLAVE_UP_LOWER_UP vtu 1500 qlen pffs, fast master mgmt-bo state
UP mode DEFAULT group default qlen 1000
    link/ether 82:54:00:03:3a:e4 brd ff:ff:ff:ff:ff:ff
...
4: mgmt-bo: BRIDGEPORT_MLTS2021_MASTER_UP_LOWER_UP vtu 1500 qlen noqueue master mgmt-br state
UP mode DEFAULT group default qlen 1000
    link/ether 82:54:00:03:3a:e4 brd ff:ff:ff:ff:ff:ff
```

- PCI device: You can use the command `lspci` to retrieve a list of devices, which allows you to quickly identify the network NICs. To retrieve detailed information about each device, use the command `lspci -v`.

Example (`lspci`)

```
# lspci
00:03.0 Ethernet controller: Intel Corporation 82564EM Gigabit Ethernet Controller (rev 03)
```

Example (`lspci -v`)

```
# lspci -v
00:03.0 Ethernet controller: Intel Corporation 82564EM Gigabit Ethernet Controller (rev 03)
Subsystem: Red Hat, Inc. QM0 Virtual Machine
Physical Slot: 3
Flags: bus master, fast dxschange, latency 0, IRQ 11
Memory at F400000 (Co-Sal, non-prefetchable) [size=1024K]
I/O ports at c000 [size=64]
Expansion ROM at F400000 [disabled] [size=512K]
Kernel driver in use: e1000
Kernel modules: e1000
```

- Linux kernel log: You can use the command `dmesg` to display kernel messages, which include most of the required information. If you save the messages to `kernel_log`, you can check the driver and link status.

Harvester places sub-NICs into the bond interfaces. In the following example, an additional bond interface named `data-bo` is created in the cluster.

```
# grep "(slave)" kernel.log (or: dmesg | grep "(slave)")
Jun 08 00:35:00 localhost kernel: mgmt-bo: (slave ensd0): Enslaving as a backup interface with an
up link
Jun 08 00:35:00 localhost kernel: mgmt-bo: (slave ensd0f0): Enslaving as a backup interface with
an up link
Jun 08 00:37:34 localhost kernel: data-bo: (slave ensd0): Enslaving as a backup interface with an
up link
Jun 08 00:37:35 localhost kernel: data-bo: (slave ensd0f1): Enslaving as a backup interface with
an up link
```

The NICs are renamed.

```
# grep "(renamed)" kernel.log
Jun 08 00:34:48 localhost kernel: tg3 0000:02:00.0 eno1: renamed from eth0 // eth0 / eno1 is not
used by Harvester
Jun 08 00:34:48 localhost kernel: tg3 0000:02:00.3 eno3: renamed from eth0 // eth0 / eno3 is not
used by Harvester
Jun 08 00:34:48 localhost kernel: tg3 0000:02:00.2 eno2: renamed from eth0 // eth0 / eno2 is not
used by Harvester
Jun 08 00:34:48 localhost kernel: tg3 0000:02:00.1 eno1: renamed from eth0 // eth0 / eno1 is not
used by Harvester
Jun 08 00:34:48 localhost kernel: i40e 0000:5d:00.0 eno4: renamed from eth0
Jun 08 00:34:48 localhost kernel: i40e 0000:0f:00.0 eno4f0: renamed from eth0
Jun 08 00:34:48 localhost kernel: i40e 0000:5d:00.1 eno4: renamed from eth0
Jun 08 00:34:48 localhost kernel: i40e 0000:0f:00.1 eno4f1: renamed from eth0
```

The NIC driver of `eno4f0(0000:5d:00.0)` is `(i40e) i40e 400Mbps Full Duplex`.

```
# grep "0000:5d:00.0" kernel.log
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0: fw 6.71.85300 opt 1.11 mss 10.54.7
[0000:1570] [1500:22fc]
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0: MAC address: 48:0f:37:04:c2:00
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0: FW LLDP is enabled
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0 eth0: NIC Link is Up, 10 Gbps Full Duplex,
Flow Control: None
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0: PCI Express: Speed 8.0GT/s Width x8
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0: Features: PF:1(s) VFs: 64 VSIs: 66 QP: 112
RSS P2:478 P2_08:1024P2_082:1024 Queue PVP VPP
Jun 08 00:34:48 localhost kernel: i40e 0000:5d:00.0 eno4: renamed from eth0
```

The enabled NICs are detected.

```
# grep "is up" kernel.log
Jun 08 00:34:47 localhost kernel: i40e 0000:5d:00.0 eth0: NIC Link is Up, 10 Gbps Full Duplex,
Flow Control: None
Jun 08 00:34:48 localhost kernel: i40e 0000:5d:00.1 eth0: NIC Link is Up, 10 Gbps Full Duplex,
Flow Control: None
Jun 08 00:34:48 localhost kernel: i40e 0000:0f:00.0 eth0: NIC Link is Up, 10 Gbps Full Duplex,
Flow Control: None
Jun 08 00:34:48 localhost kernel: i40e 0000:0f:00.1 eth0: NIC Link is Up, 10 Gbps Full Duplex,
Flow Control: None
```

Enable Maintenance Mode

1. (Optional) Stop VMs that cannot or must not be migrated.
 2. Enable **maintenance mode** on the target node to automatically migrate all VMs to other nodes.
- Wait for everything to become ready, and then repeat the steps in the **Pre-Replacement Checks** section.
 - Manually stop a VM in the following situations:
 - The VM fails to migrate.
 - The VM has selectors that prevent it from migrating to other nodes.
 - The VM has special hardware (for example, PCI passthrough or vGPU) that prevent it from migrating to other nodes.

(Optional) Update the Network Config

There are one or more **Network Config** objects under every **Cluster Network** on Harvester. Each **Network Config** is backed by a **VlanConfig** CSD object.

ⓘ IMPORTANT

Updating the **Network Config** is **required** if the new NICs will be placed in different physical slots or will have different `ipnet` parameters.

1. Check the node.

When a Harvester cluster node belongs to a **Network Config**, the **Node** object has a label with the key `network.harvesterhci.io/vlanconfig`.

Example

```
apiVersion: v1
kind: Node
metadata:
  labels:
    network.harvesterhci.io/vlanconfig: vlan123
```

2. Remove this node from the **Network Config**.

When the new NICs are placed in different slots, you must change the **Network Config** to exclude this node. You can delete the **VlanConfig** if the **Network Config** object selects only this node from `nodeSelector`.

Example

```
apiVersion: network.harvesterhci.io/v1beta1
kind: VLANConfig
spec:
  clusterNetwork: data
  nodeNetwork:
    kubeNetwork: node123 // select one or more nodes
  uplink:
    bandwidth:
      min: 500
      mode: RST_3rd
    linkAttributes:
      etw: 1000
      toQem: -1
      nic:
        - enp0s1
        - enp0s2
```

When VMs are still running on an affected node, the network webhook returns an error.

3. Check the `Node` object.

Depending on the situation, either the label `network.harvesterhci.io/vlanconfig` changes or is removed.

4. Check the `VLANStatus` object.

Depending on the situation, either the status of the `VLANStatus` object's `ready` condition changes to `"True"` or the object is deleted.

Example

```
apiVersion: network.harvesterhci.io/v1beta1
kind: VLANStatus
networks:
  -
  status:
    clusterNetwork: data
    conditions:
      - lastUpdateTime: "2008-03-03T18:32:41Z"
      status: "True"
      type: ready
    linkSelector: public
    localArea:
      - cidr: 10.100.100.0/24
      vlanID: 2013
      node: node123
      vlanConfig: vlan123
```

(Optional) Drain the Node

You may find that some Longhorn replicas remain active on the node even after completing the previously outlined procedures.

1. Drain the node. (This is optional in Harvester.)

- Scenario 1: The `multiReplica` value of all volumes is `3`, which means that each Longhorn volume has three active replicas.

The Longhorn Engine recognizes that it can no longer communicate with the replica on the drained node, and then marks that replica as failed. None of the replicas hold any special significance to Longhorn so it functions as long as it can communicate with at least one replica.

- Scenario 2: Some Longhorn volumes have fewer than three active replicas, or you manually attached volumes using the Harvester UI or Longhorn UI.

You must manually detach the replicas or move them to other nodes, and then [drain the node](#) using the command `kubectl drain -i --ignore-daemonsets --node name`. The option `--ignore-daemonsets` is required because Longhorn deploys daemons such as Longhorn Manager, Longhorn CSI plugin, and Longhorn Engine image.

Replicas running on the node are stopped and marked as `Failed`. Longhorn Engine processes running on the node are migrated with the pod to other nodes. Once the node is fully drained, no replicas or engine processes should remain running on the node.

2. Replenish replicas.

After a node is shut down, Longhorn does not start rebuilding the replicas on other nodes until the `replicas-replenishment-wait-interval` (default value: 600 seconds) is reached. If the node comes back online before the wait interval value is reached, Longhorn reuses the replicas. Otherwise, Longhorn rebuilds the replicas on another node.

During system maintenance, you can modify the `replicas-replenishment-wait-interval` value using the [embedded Longhorn UI](#) to enable faster replica rebuilding.

Harvester v1.3.0 uses Longhorn v1.6.0, while Harvester v1.2.1 uses Longhorn v1.4.3.

Replace the NICs

1. Shut the node down.

2. Replace the NICs.

3. Restart the node.

4. [Collect information](#) about the current network configuration and status.

If you observe any abnormalities, generate a [support bundle](#) for troubleshooting purposes.

(Optional) Update the Network Config Again

⚠ IMPORTANT
Updating the `Network Config` is [required](#) if the new NICs will be placed in different physical slots.

1. Add the node to the `Network Config`.

You must create a new `Network Config` or change the `Network Config` to include this node.

2. Check the `Node` object.

The label `network.harvesterhci.io/vlanconfig` reflects the specific `Network Config` used.

3. Check the `VLANStatus` object.

The status of the `VLANStatus` object's `ready` condition changes to `"True"`.

Disable Maintenance Mode

1. Wait for the node to be moved back to the cluster.

2. Disable maintenance mode.

3. (Optional) Start the VMs that you manually stopped.

4. (Optional) Manually [migrate VMs](#) to this node.

Troubleshooting

Harvester uses multiple network-related pods and CRDs. When troubleshooting, check the pod logs and the status of CRD objects.

Pods:

```
# kubectl get pods -n harvester-system
NAME                                READY   STATUS    RESTARTS   AGE
harvester-network-controller-crd    1/1     Running   0 (50m ago)   3d0h //
Network controller agent daemonSet, deployed in each node
harvester-network-controller-manager-886c4cd74-rc1f   1/1     Running   0 (50m ago)   3d0h //
Network controller
harvester-network-watchdog-56b877df6f-x42dp           1/1     Running   0 (50m ago)   3d0h
```

CRDs:

```
clusterNetworks.network.harvesterhci.io
linkSelectors.network.harvesterhci.io
vlanconfigs.network.harvesterhci.io
vlanStatuses.network.harvesterhci.io
```